



LIGHT MATE

THINK • PLAY • CONNECT

Team Light Mate

CPEG/ELEG499, University of Delaware, Spring 2026

The Team

Anas

CYBER

Software
Development

Omar

CPEG

Schematic PCB Design
and CAD

James

ELEG

PCB Layout and
Design



Why

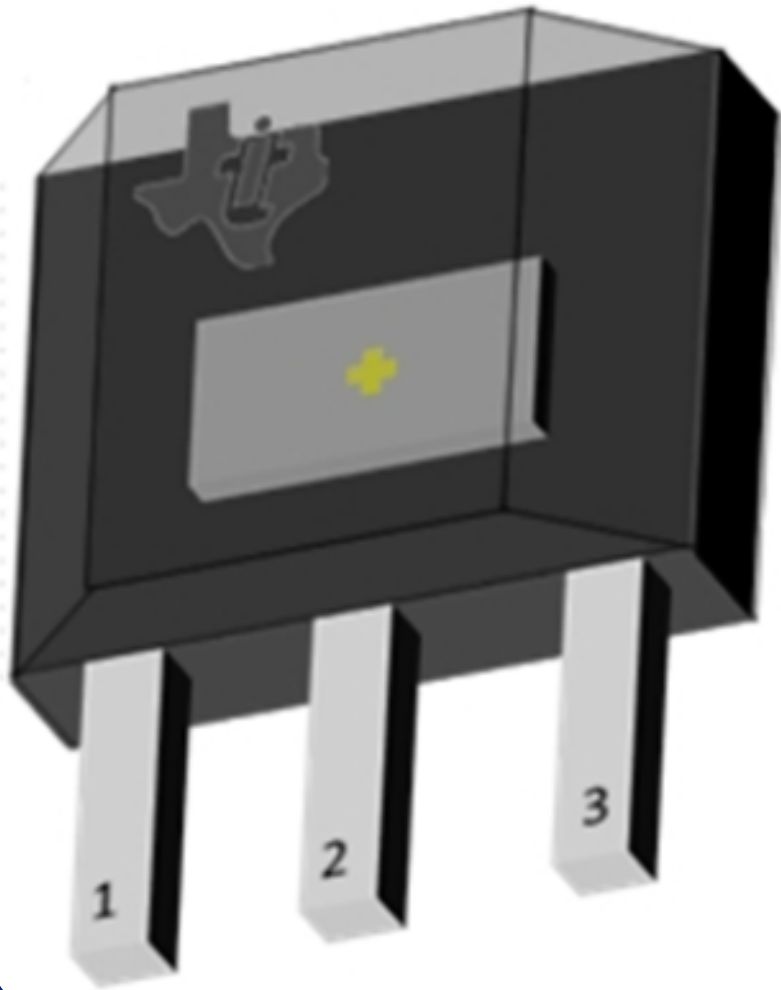
The Smart Chessboard was designed to enhance **tournament-level training** through digital analysis and real-board experiences, supporting tactics and even blindfold chess practice.

Initial Concept

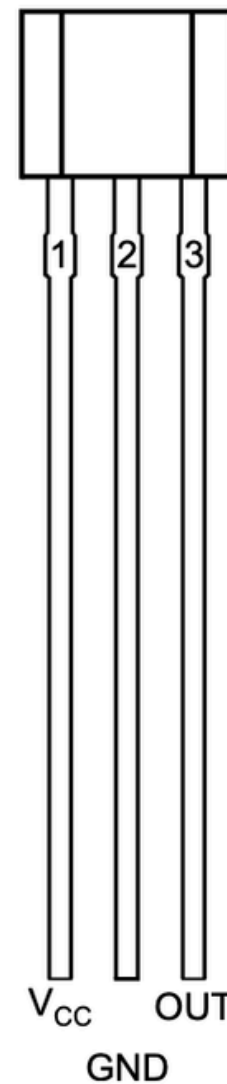
The **initial concept** involved using RFID technology for each chess piece. However, we found this may lead to **wiring complexity** and slow detection, causing inaccuracies in identification.



TO-92



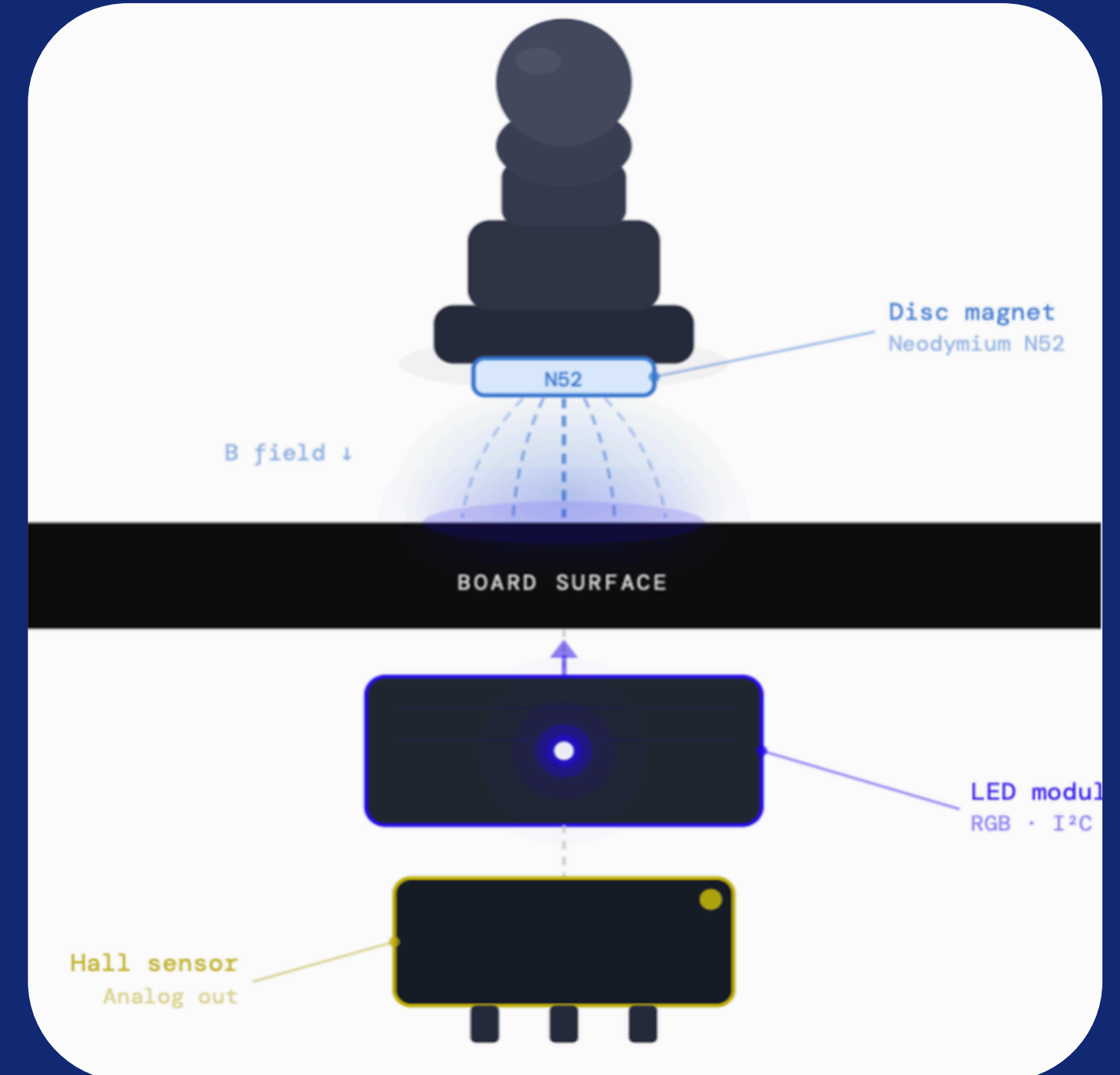
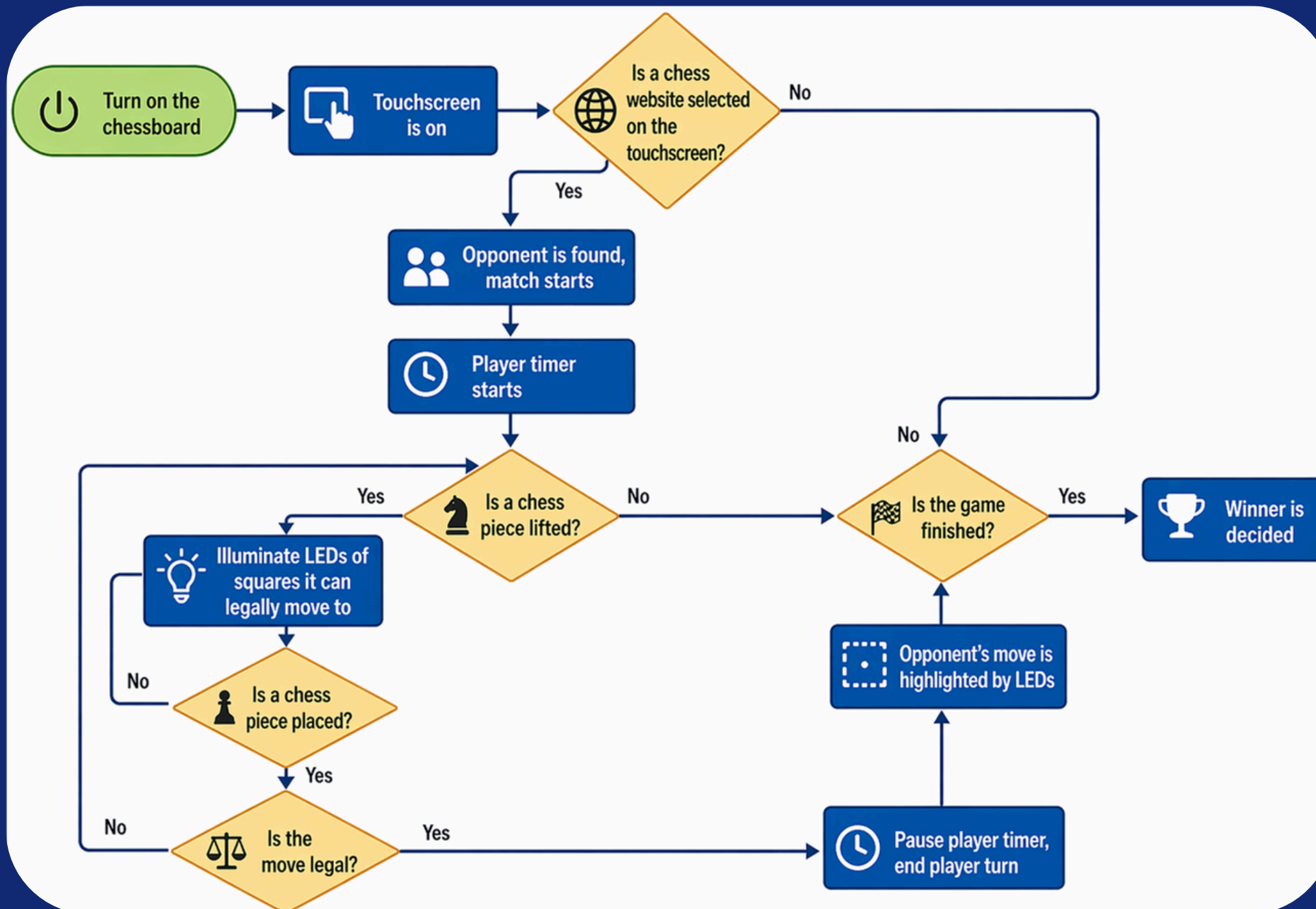
LPG Package
3-Pin TO-92
Top View

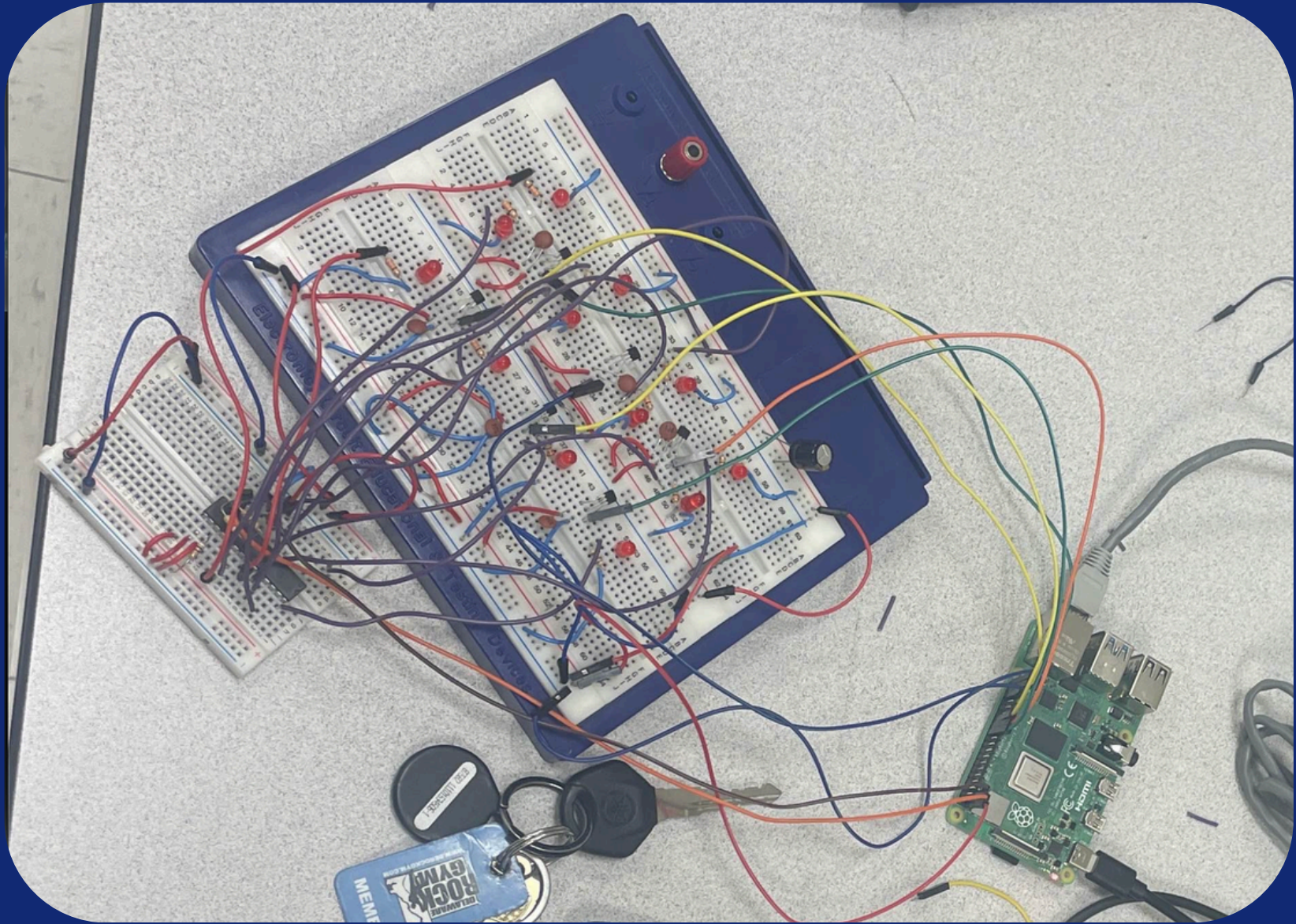


Why Hall Sensors

Hall sensors provide **fast and reliable detection** for our chessboard, enabling accurate piece identification at a lower cost while ensuring scalability for all 64 squares.

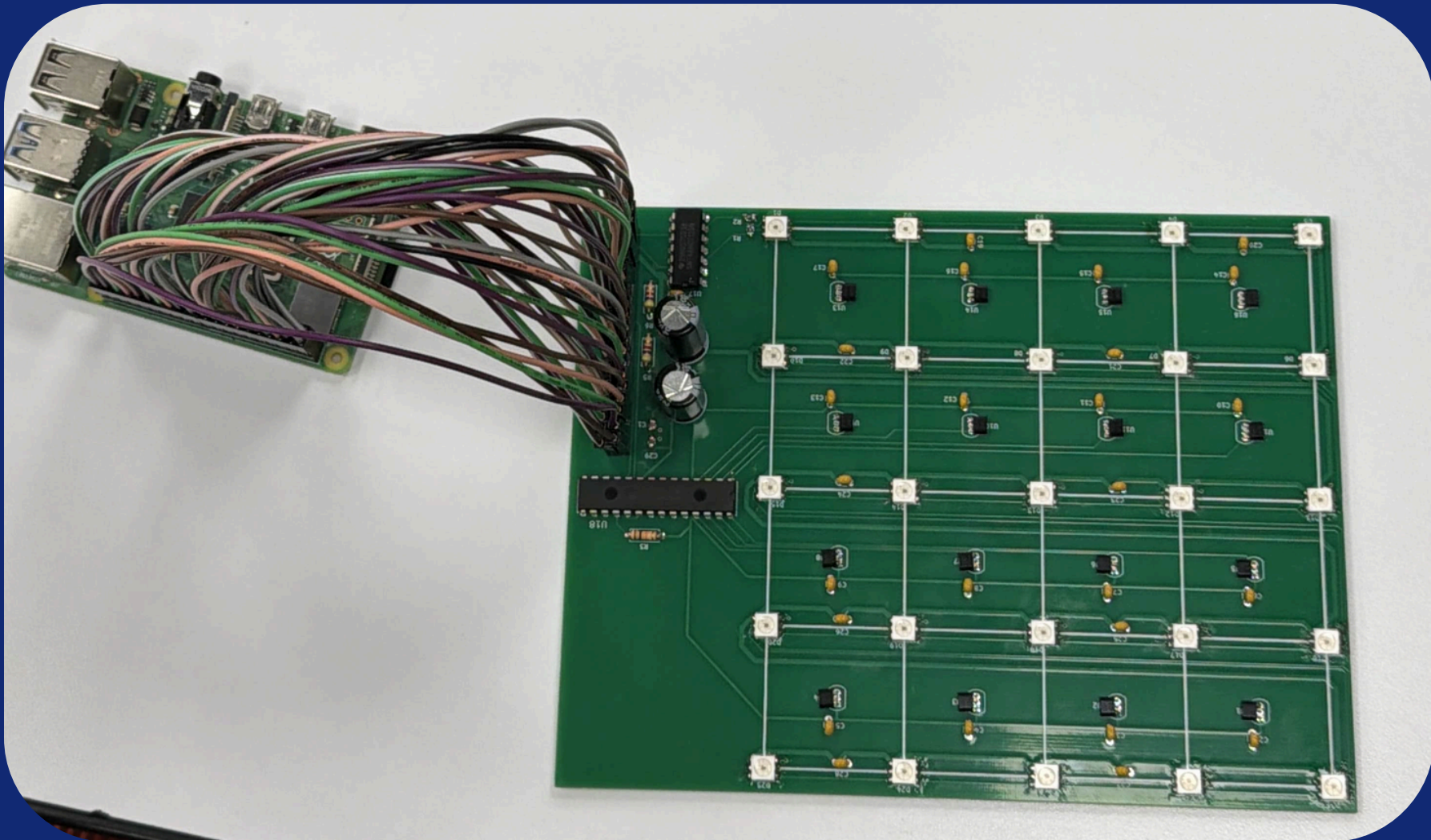
System Architecture





Fall 2025 Prototyping

In this phase, we tested the sensors and the LED chain, ensuring proper wiring layout for optimal performance and reliability in the our project.



Fall 2025 Results

A **4×4 PCB prototype** was successfully assembled with all sensors soldered, LEDs functioning correctly, and the Raspberry Pi connected, validating our logic for the chessboard project.

Engineering Constraints

Key factors impacting our design process

Size & Weight

Compact circuit fitting all components.

Timing Requirements

Poll sensors quickly enough to detect real time piece movement.

Power Management

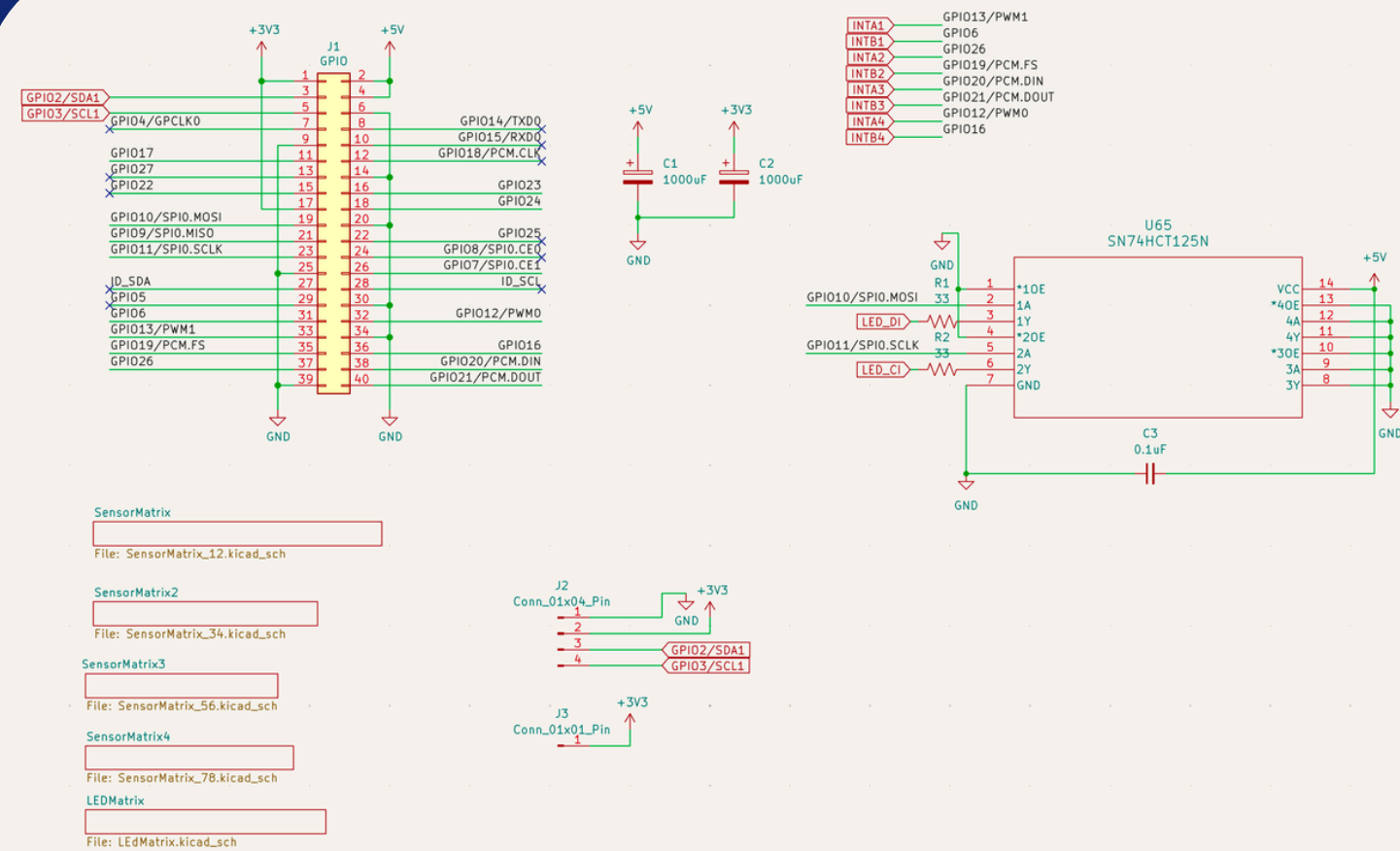
Level shifter used to drive LEDs with lower brightness to avoid overloading Pi power pins.

Cost Considerations

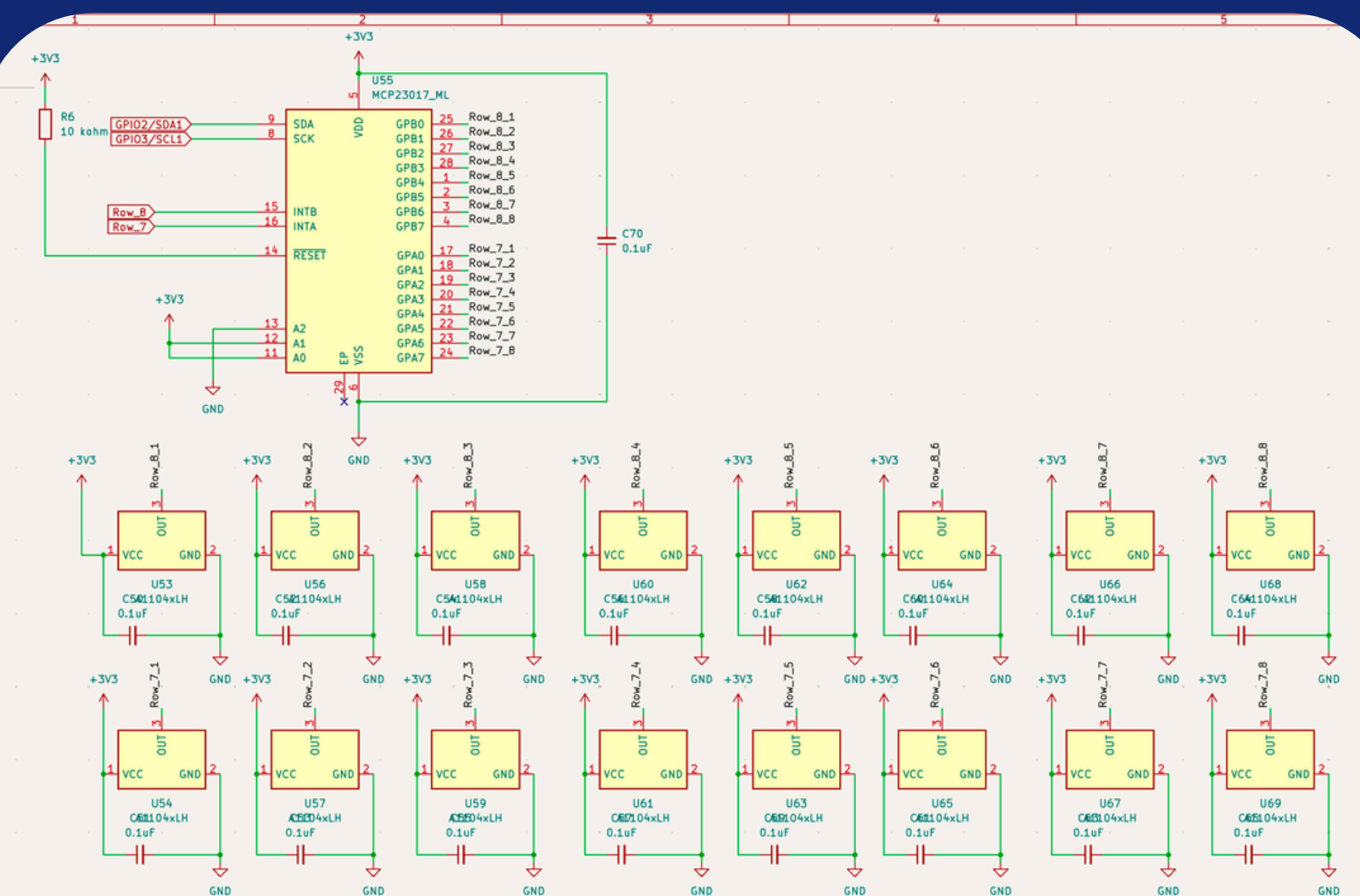
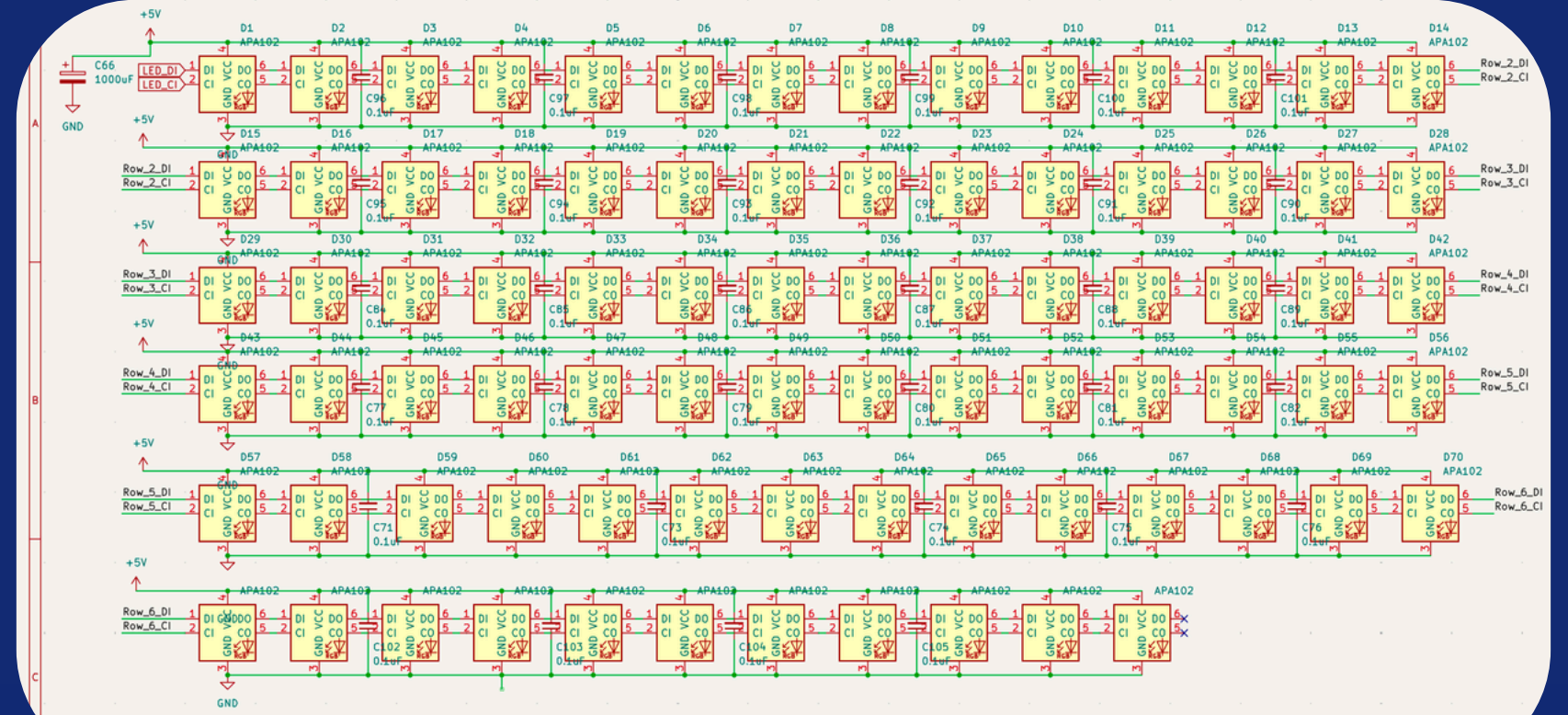
All constraints were optimized to reduce cost per square and fit within budget limits.

Scaling to 8x8

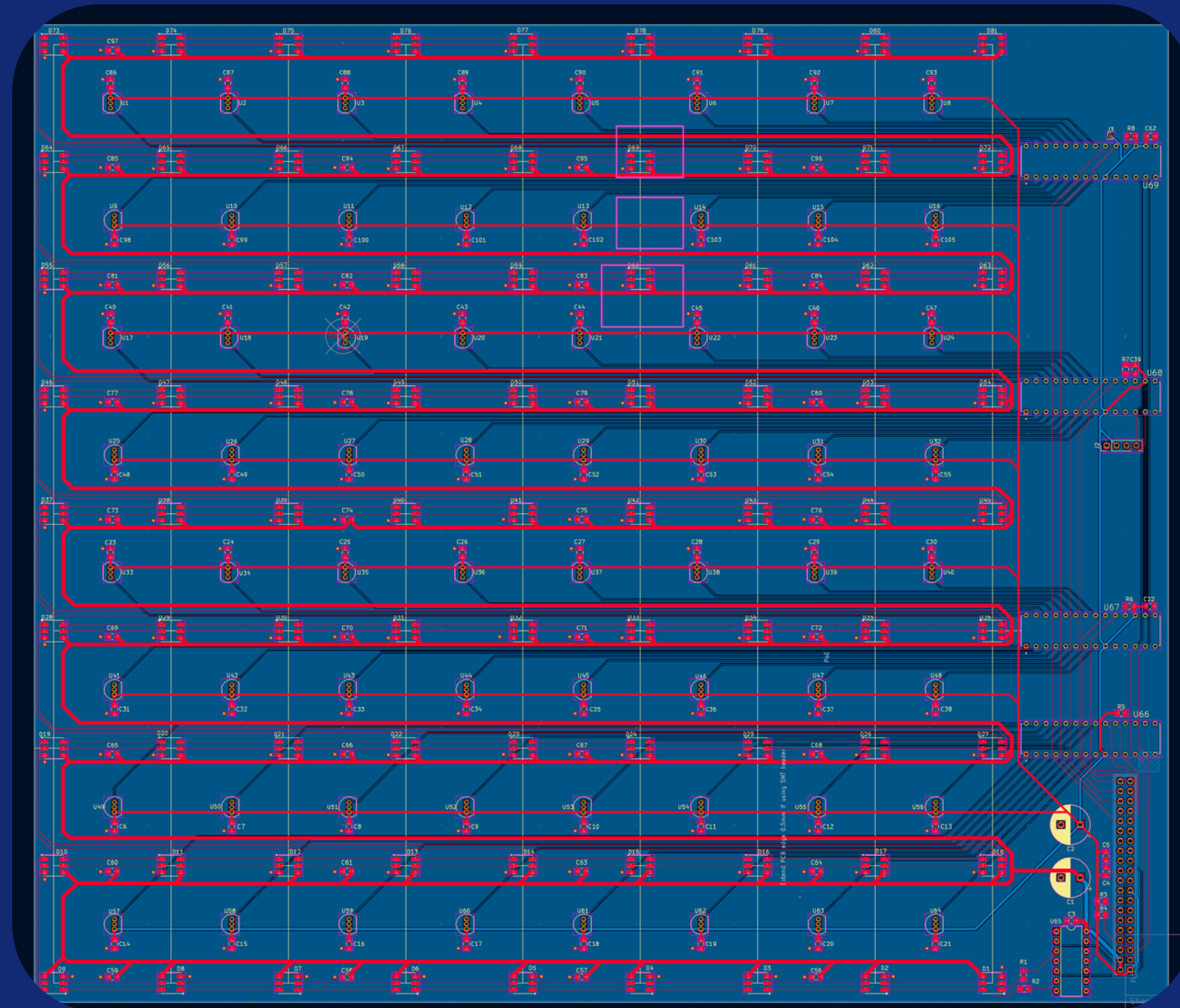
Transitioning from a 4x4 to a full 8x8 board involved complex **routing challenges**, including managing power rails, accommodating 64 sensors, and integrating 81 LEDs efficiently.



<https://pinout.xyz/>



Scaling to 8x8



Challenges Encountered

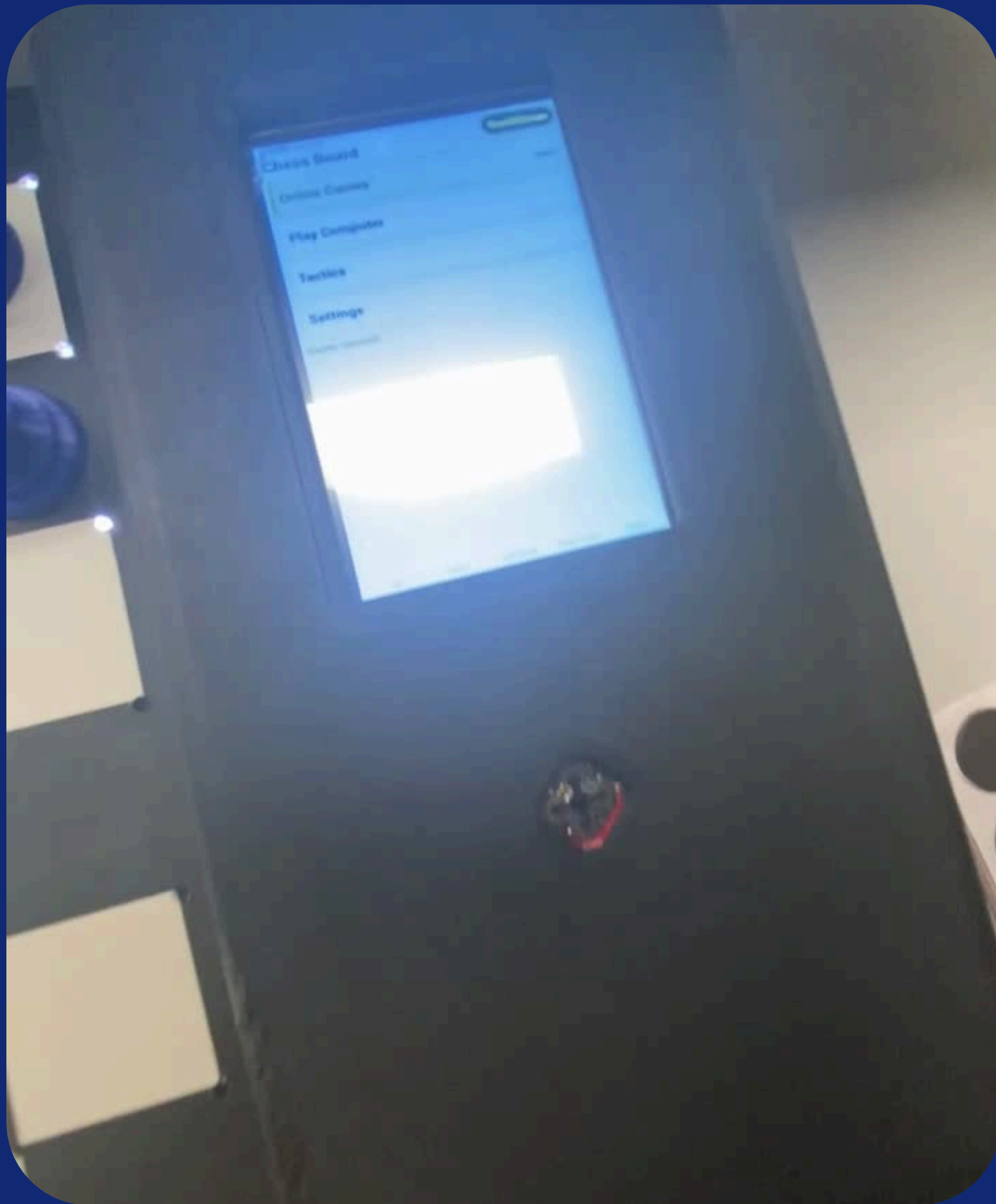
- During the soldering of our first 8x8 PCB, we didn't have the right process to test the connections and pi while attaching components.
 - As a result, the pi would not boot up to be written with everything soldered.
 - We made changes to fix this, such as inspecting each solder joint and resoldering. We ended up shorting the powerline and fried our board.
- Instead of attempting to salvage the situation, we decided to move onto a new board.
 - We tested the resistivity of the bare board, used the continuity function of multimeter to test every connection as we solder, and occasional test boots of the pi.
 - Also dealt with issues regarding the LEDs. Data signals transferring to each other due to weak soldering connections and faulty LEDs.

CAD Design



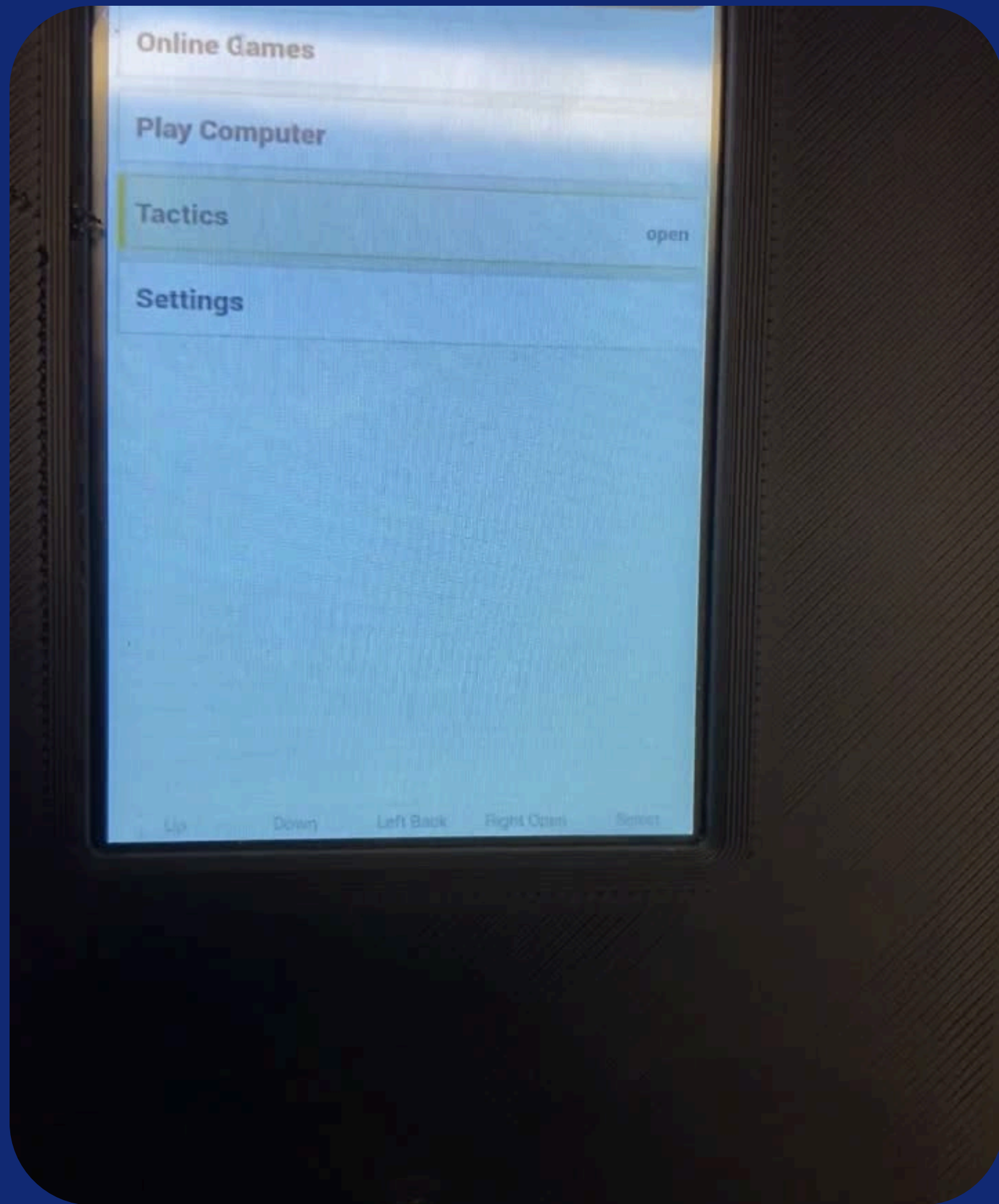
Future Work

- Replace Raspberry Pi 4 with a Compute Module integrated into a custom PCB to reduce wiring and enable a smaller, more compact enclosure.
- Add mobile/web interface for game tracking, remote play, and cloud synchronization.
- Implement battery power and optimized power management for a fully portable system.
- Increase Screen size and implement touch interface.
- Use better materials and improve durability of Enclosure.



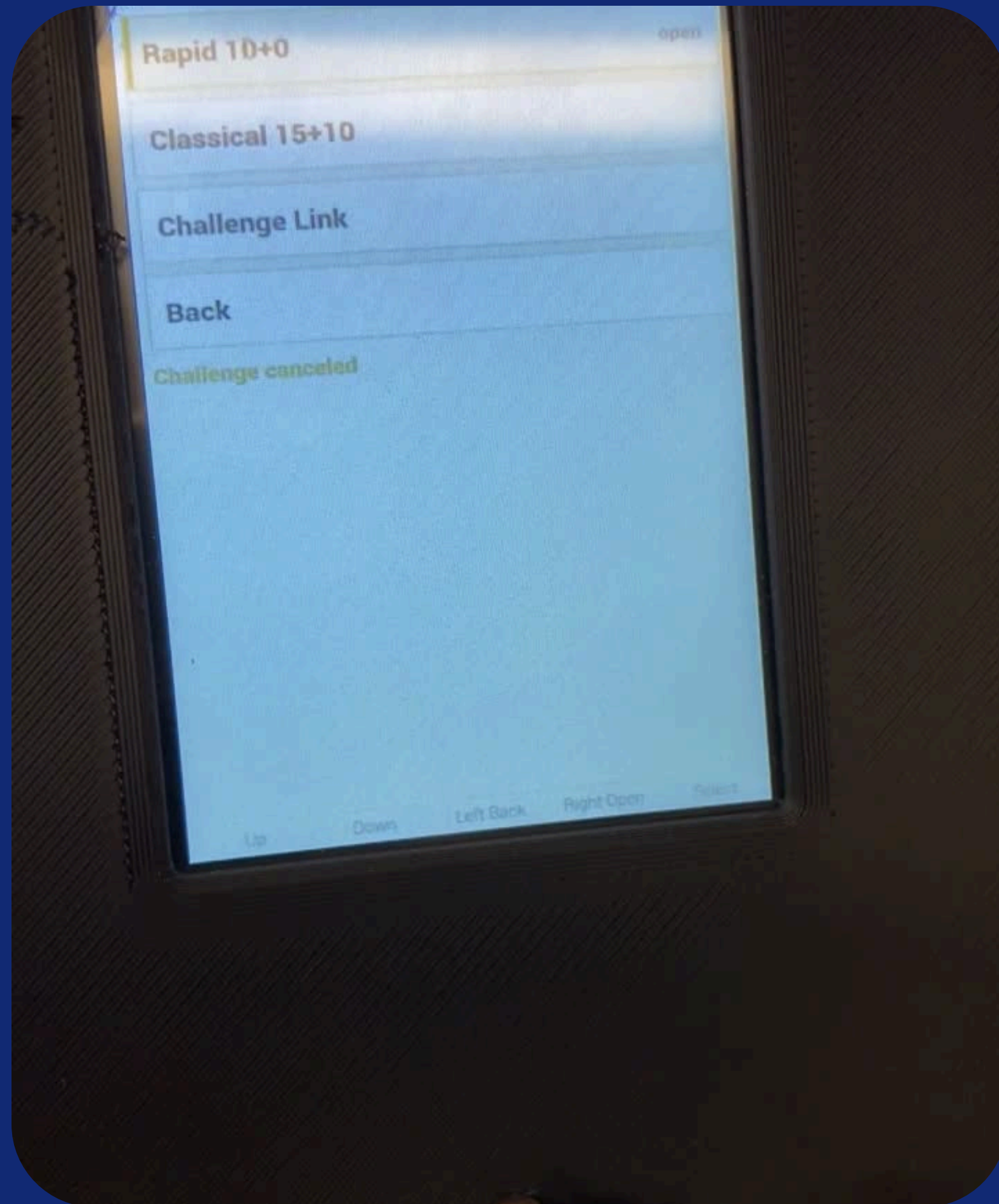
Video Demo

Online Game



Video Demo

Puzzle



Video Demo

Settings

Questions?



Github

Code, demo videos & additional Project
information

